

Vision

Our vision is to revolutionize agriculture by empowering farmers through our Smart Agriculture Solution, a paradigm-shifting innovation delivering data-driven, economical, and sustainable solutions.

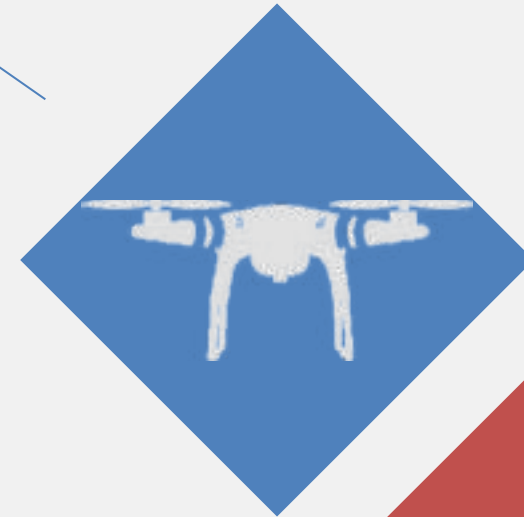
Problem Statement:

- **Multifaceted Agricultural Challenges:** Addressing issues such as pest infestations, crop diseases, and efficient livestock management.
- **Need for Advanced Technology:** Conventional methods are laborious and ineffective, necessitating the development of an integrated solution.
- **Sensor-Equipped Farm Monitoring Drone System:** Aimed at bolstering agricultural productivity and sustainability.
- **Integration of Machine Learning Algorithms:** For precise pest detection, disease identification, and crop health insights.
- **User-Friendly Farm Monitoring Software:** Ensuring accessibility and ease of use for farmers and agricultural workers..

Key Objectives

Quality Crop Monitoring

Utilizing innovative UAV systems for accurate, real-time crop development and wellness tracking.



Soil Condition Measurement

Utilizing sensors for measuring soil quality to provide comprehensive data to farmers at their fingertips



Smooth Data Collection and Transfer

Establishing an IoT infrastructure for seamless data sharing, enabling efficient data analysis and transfer



Empowering Farmers

Implementing innovative technologies for data-driven, efficient, and sustainable smart agriculture



Sustainability and Efficiency

Optimizing components with 3D designs, using a highly affordable multispectral camera for improved data measurement.

Research Survey:

- Delavarpour et al. advocate integrating cutting-edge technology with UAVs in precision agriculture to enhance output and reduce costs, emphasizing the need for improved sensing technologies and methodology integration to overcome resolution limitations and weather dependency.
- Song et al. assess four UAV-based granular fertilizer spreaders, with GFS-D showing superior performance, particularly in accurate variable-rate fertilization. Future work will focus on developing an autonomous variable-rate control system for precise site-specific fertilizer management.
- Rangappa et al. introduce LEDNet, which employs low-power LED communication and UAVs for high-precision data extraction in precision agriculture. Future studies will explore multi-color LEDs and higher-resolution cameras to enhance precision, alongside efforts to extend UAV endurance and improve power usage synchronization techniques.
- Sharma et al. present an IoT-based fuzzy-logic pest prediction system for rice and millet crops, highlighting the impact of meteorological conditions on pest occurrence. Future research aims to optimize fuzzy logic using genetic algorithms on edge devices to generate Fog-based recommendations, improving the accuracy of farmers' preventive measures.

Proposed Solution:

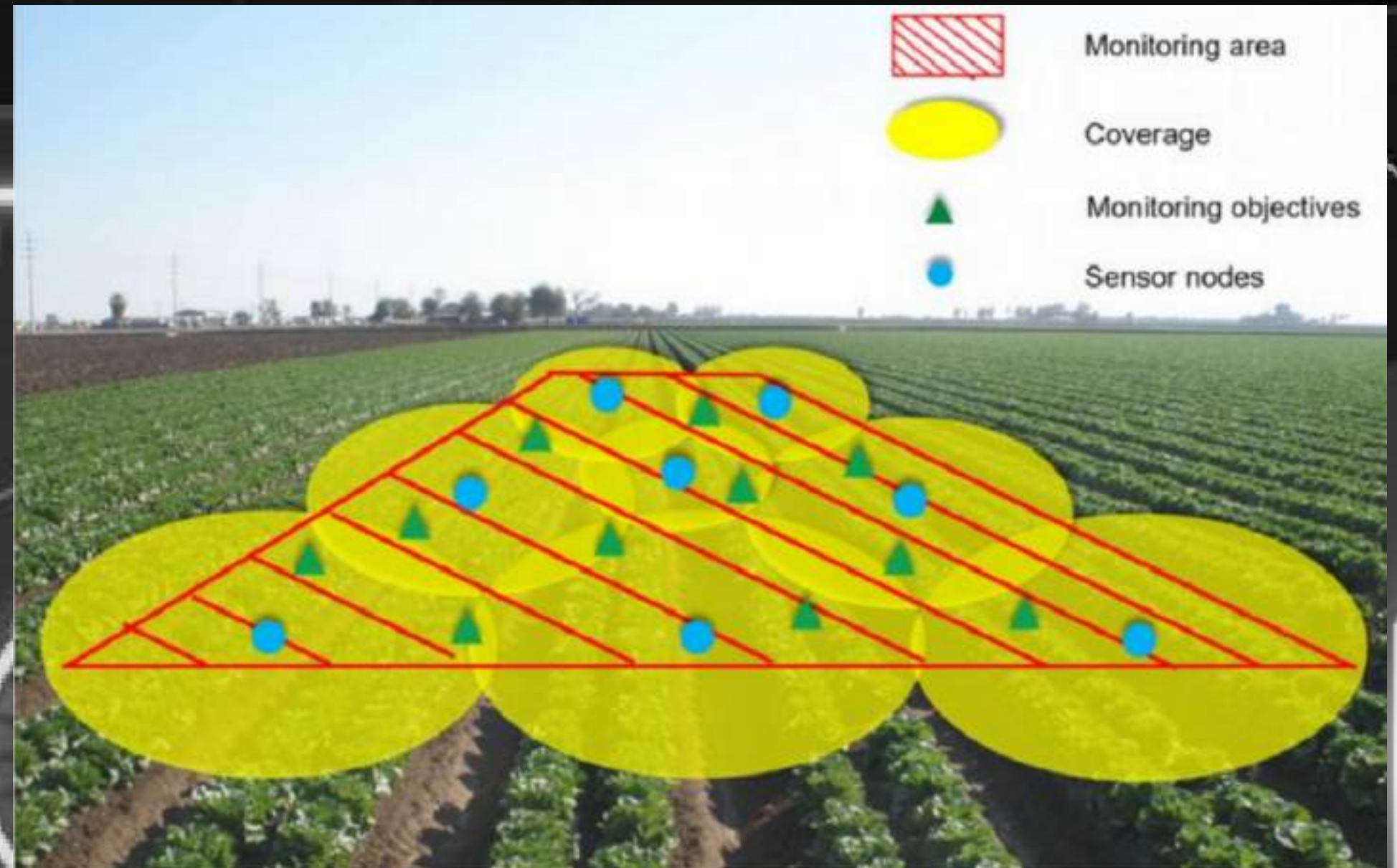
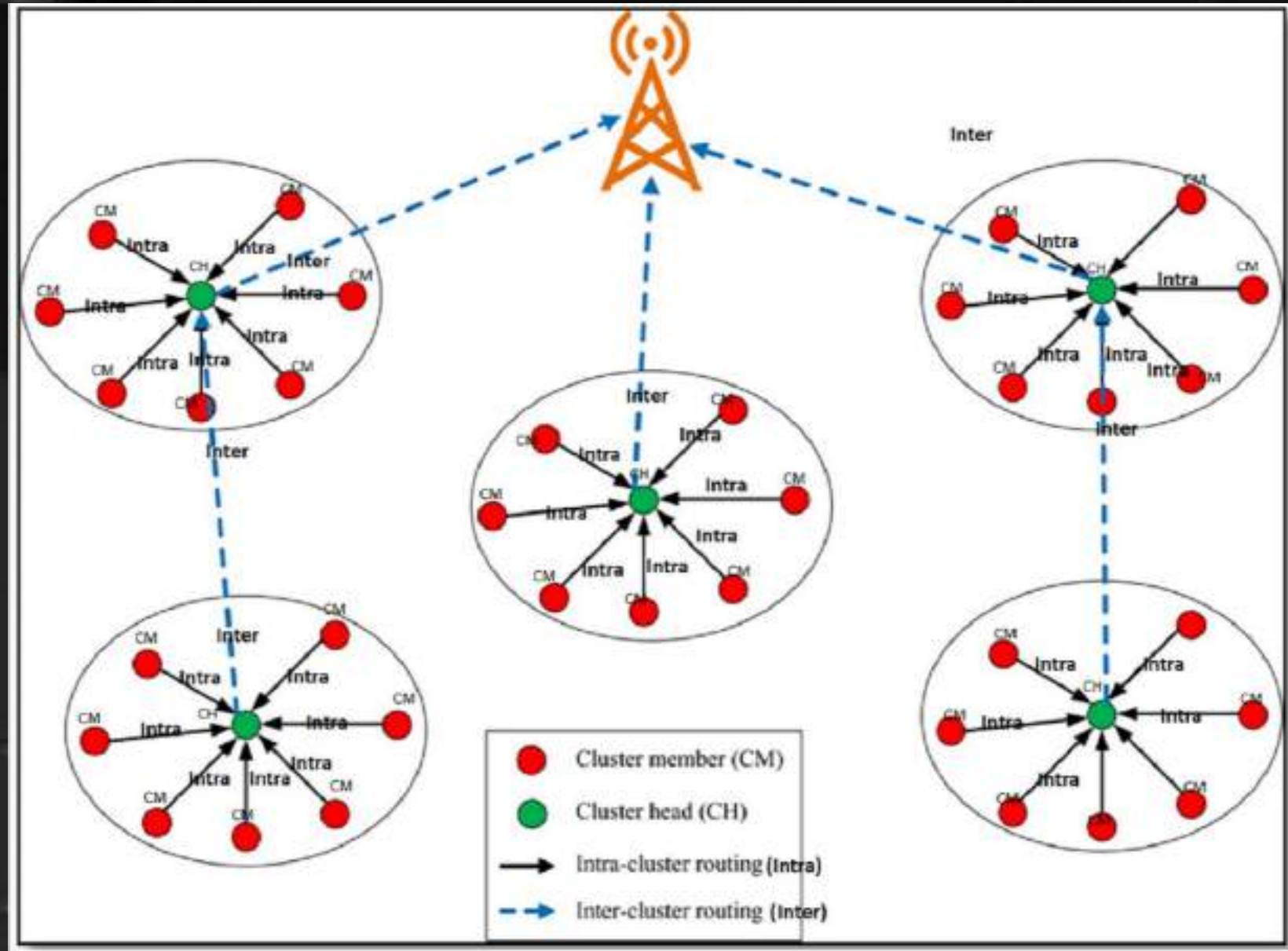
- **Drone Development:** Design and build a sensor-equipped farm monitoring drone with various sensors like cameras and multispectral imaging systems for data collection.
- **Data Collection and Processing:** Deploy the drone to gather data on crop health, pest infestations, and other parameters, processing it onboard or transmitting it to a ground station.
- **Machine Learning for Pest Detection:** Develop machine learning algorithms to detect pests and identify crop diseases from aerial imagery.
- **Crop Recognition System:** Create a crop recognition system using machine learning to classify crops and assess their health and growth stages.
- **Farm Monitoring Software:** Develop user-friendly software for farmers to access and analyze drone-collected data, featuring visualization tools and customizable alerts.
- **Integration and Testing:** Integrate components into a unified system and conduct extensive testing for reliability and usability under various conditions.
- **Deployment and Training:** Deploy the system to farms for real-world testing, providing training to farmers and workers on its use and functionalities.



OUR UAV SYSTEM



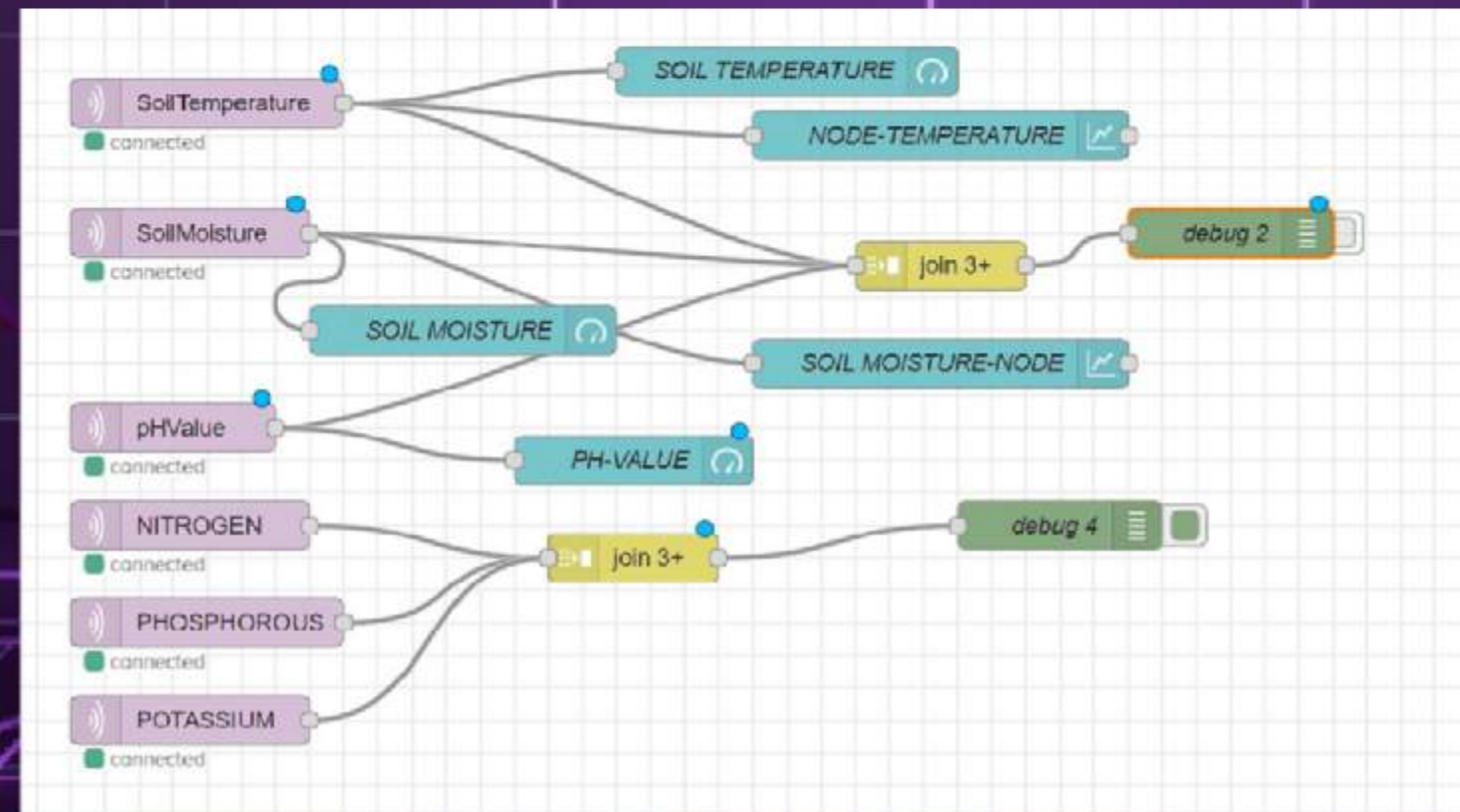
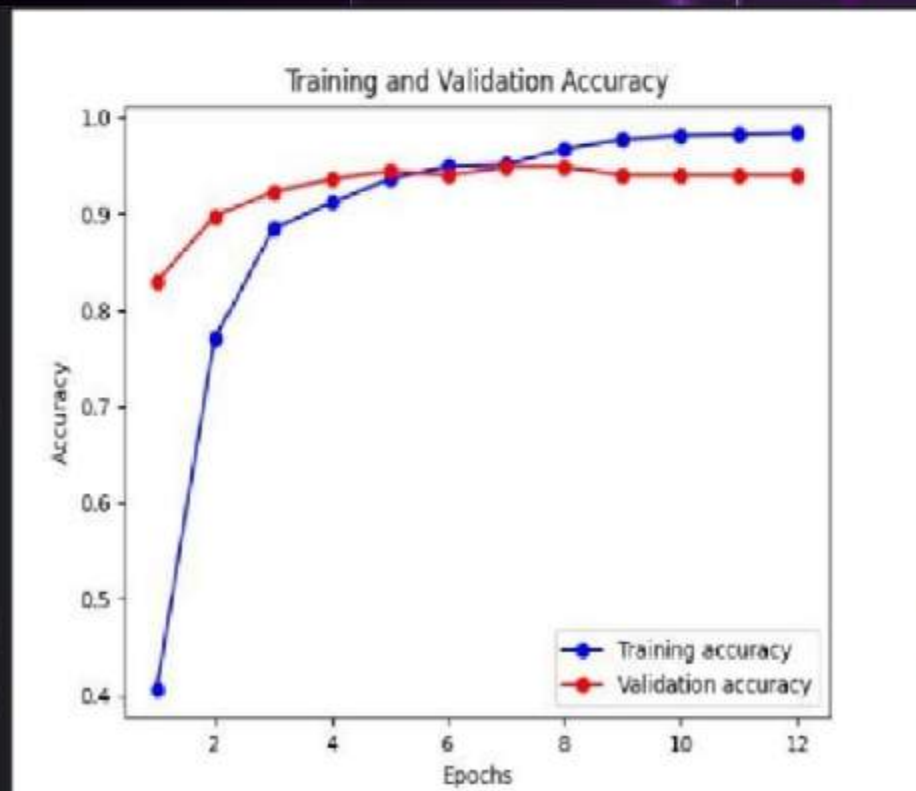
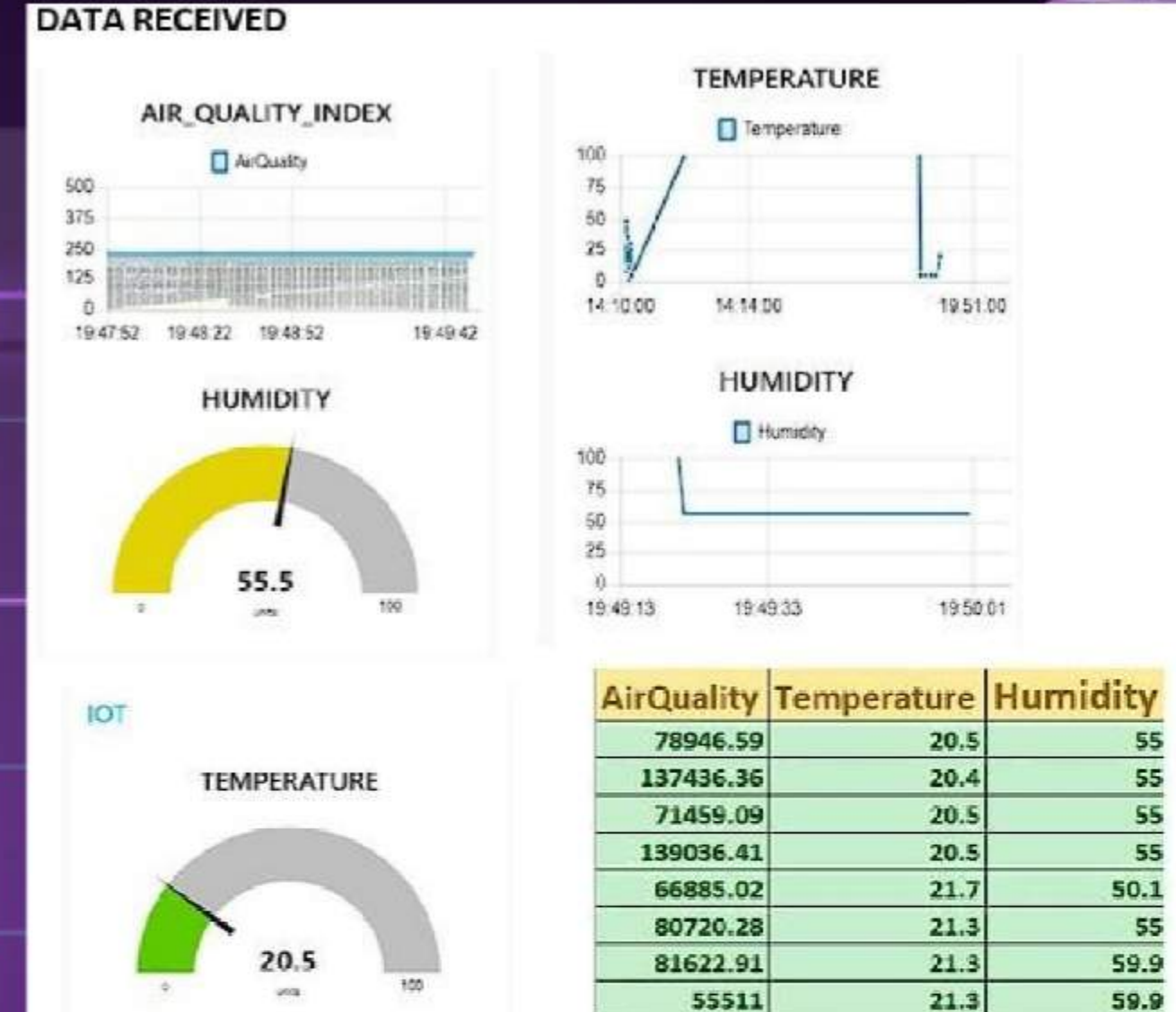
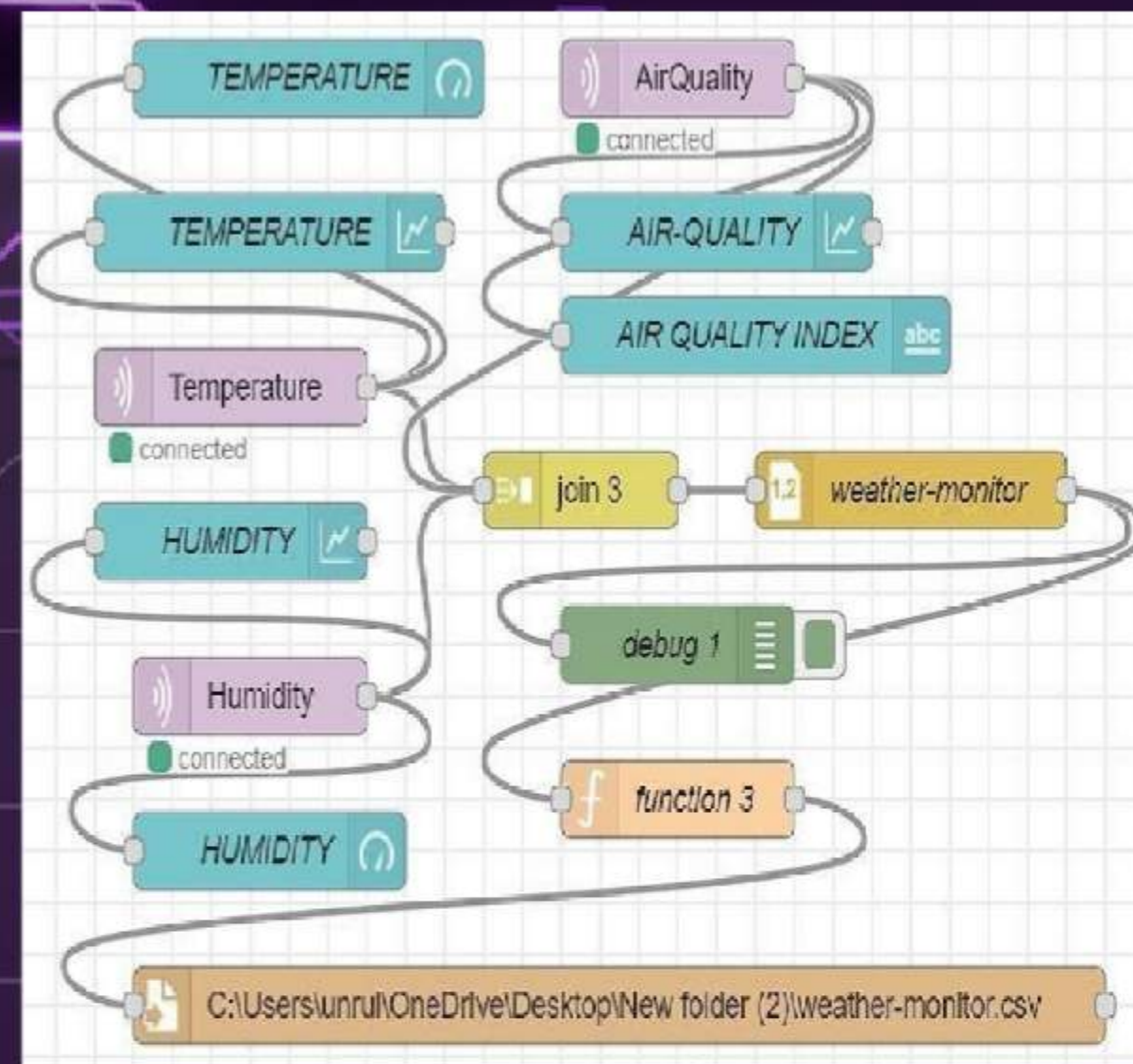
OUR AUTONOMOUS FLIGHT TEST



DEW NODE AND THE DATA
CLUSTERING INTO MAIN
STATION

CHARGING
INTERMEDI
ATE





A COMPLETE
COMPREHENSIVE
WORKFLOW AND
DATA COLLECTION
OF OUR SYSTEM
AND THE COMPLETE



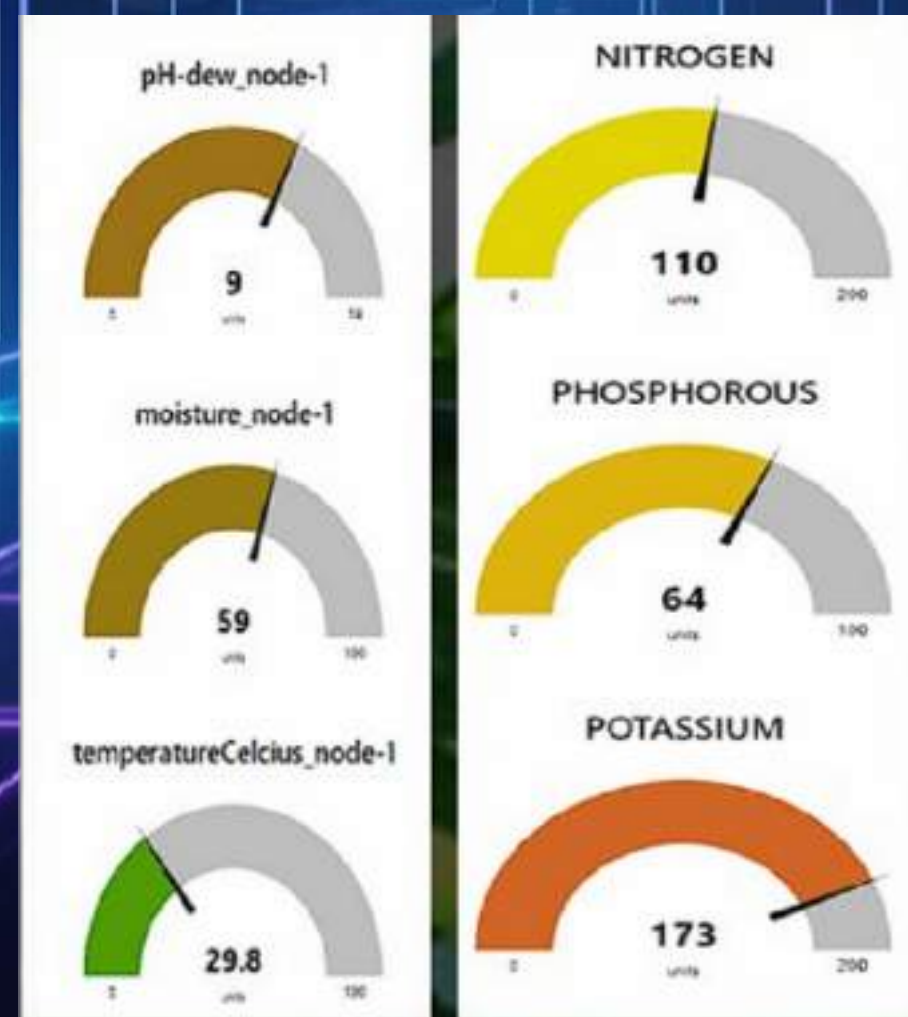
WET SOIL TEST



SENSOR ENABLED
DEW
NODE

TIME	NITROGEN[PPM]	PHOSPHOROUS[PPM]	POTASSIUM[PPM]	PH	MOISTURE[%]
09.6	114	72	180	8	63
09.6	112	70	170	8	63
09.6	113	70	181	8	62
09.8	114	72	187	8	63
09.7	112	72	188	8	62
09.8	116	70	180	7	60
09.7	114	70	187	7	60
09.7	113	72	182	7	63
09.7	112	70	180	8	63
09.7	112	70	188	8	63
09.7	112	70	182	8	63
09.7	112	70	182	7	63
09.7	112	70	187	8	63
09.7	112	70	182	8	63
09.7	112	70	187	8	63
09.7	112	70	187	8	63
09.7	112	70	187	8	63
09.7	112	70	182	8	63
09.6	114	70	181	8	63
09.6	114	70	187	8	63
09.6	112	70	182	8	63
09.6	113	70	181	8	63
09.6	113	70	182	8	63
09.6	113	70	182	8	63
09.6	113	70	182	8	63

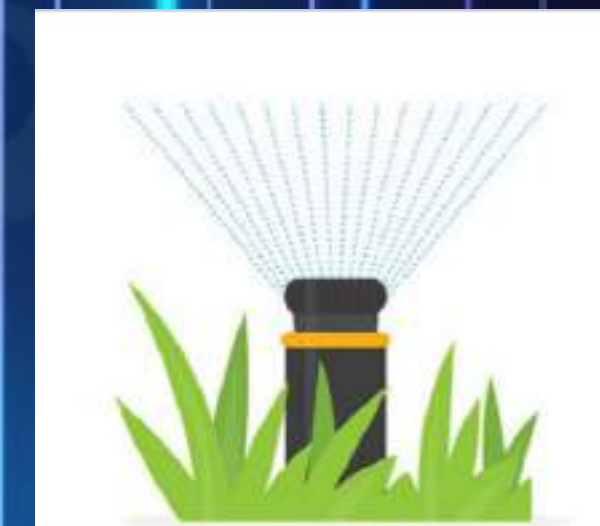
DATA COLLECTION,
SENSOR ACTUATION,
DATA TRANSFER,
SPRINKLER SYSTEM



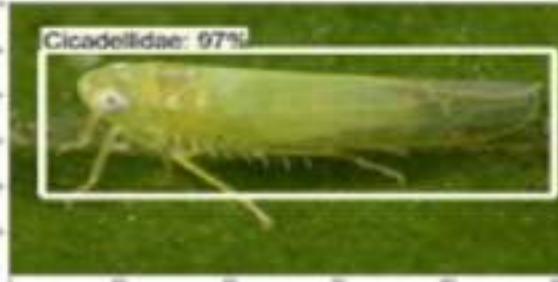
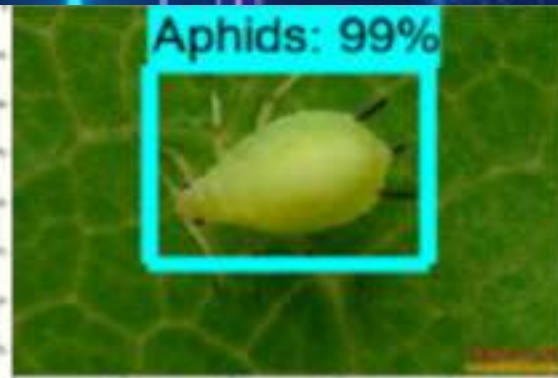
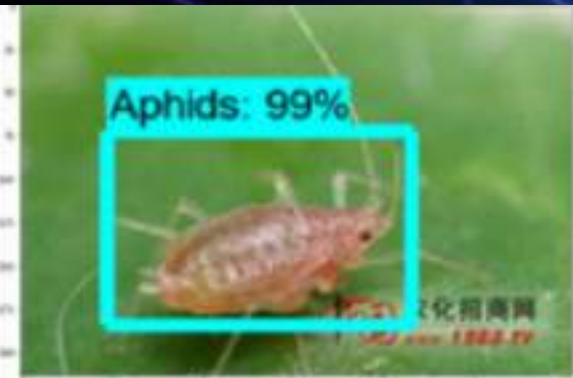
SOIL DATA COLLECTED
AND TRANSFERRED



DRONE COLLECTS AND
TRANSMITS DATA



SPRINKLER AUTOMATION
ENABLED FOR PLANTS

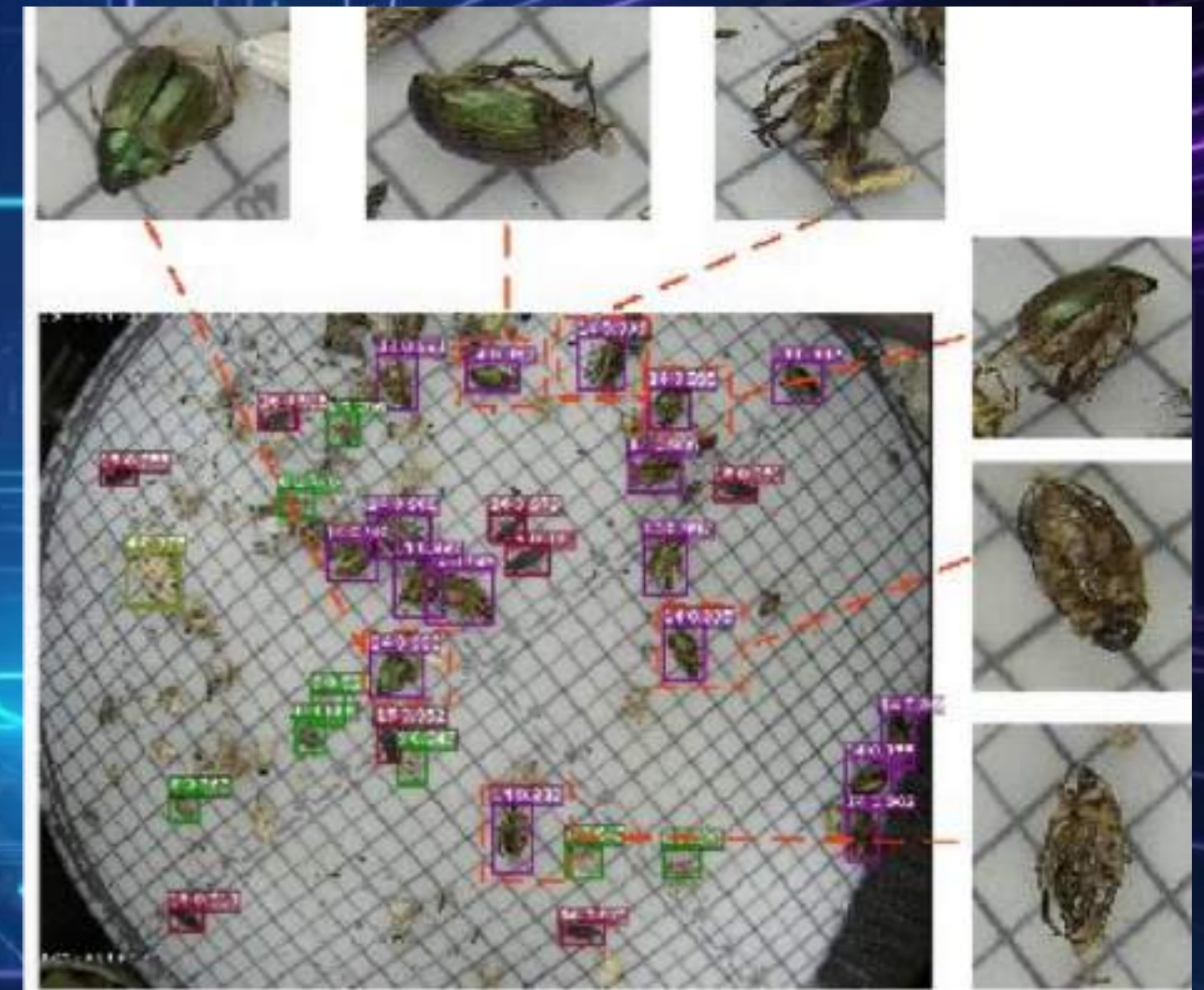


PEST DETECTION AND ANALYSIS

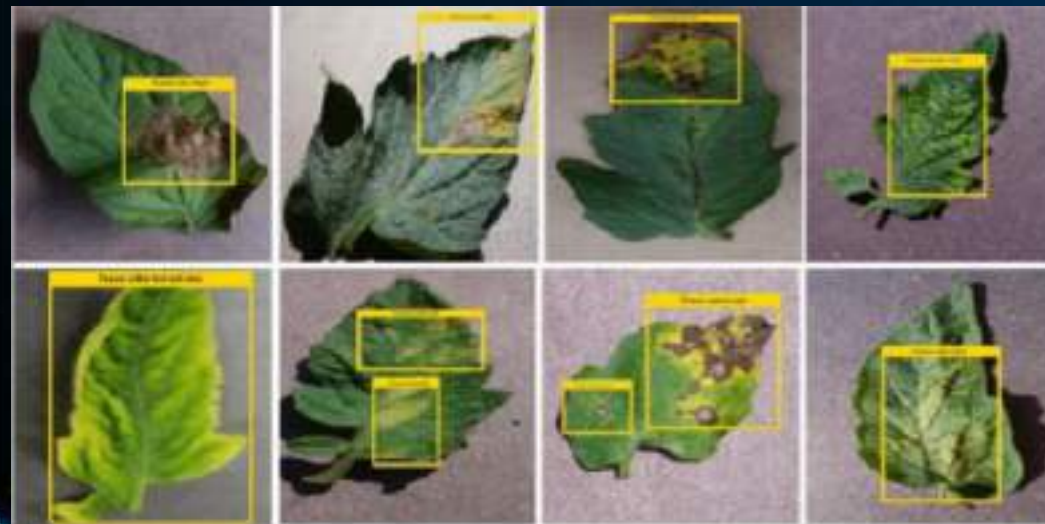
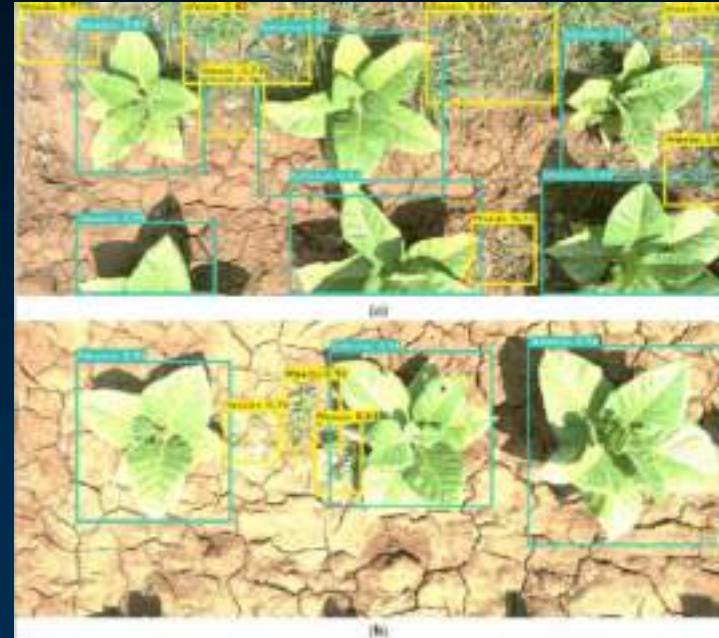
PREDICT THE DIRECTION
OF THEIR INFESTATION
AND POISONING

DETERMINE WAYS TO
PREVENT INFESTATION
AND SOLUTION OF
ELIMINATION
DEPENDING ON PEST
COVERAGE

MOBILE APP ALARM FOR INCREASED
INFESTATION



CROP AND WEED DETECTION

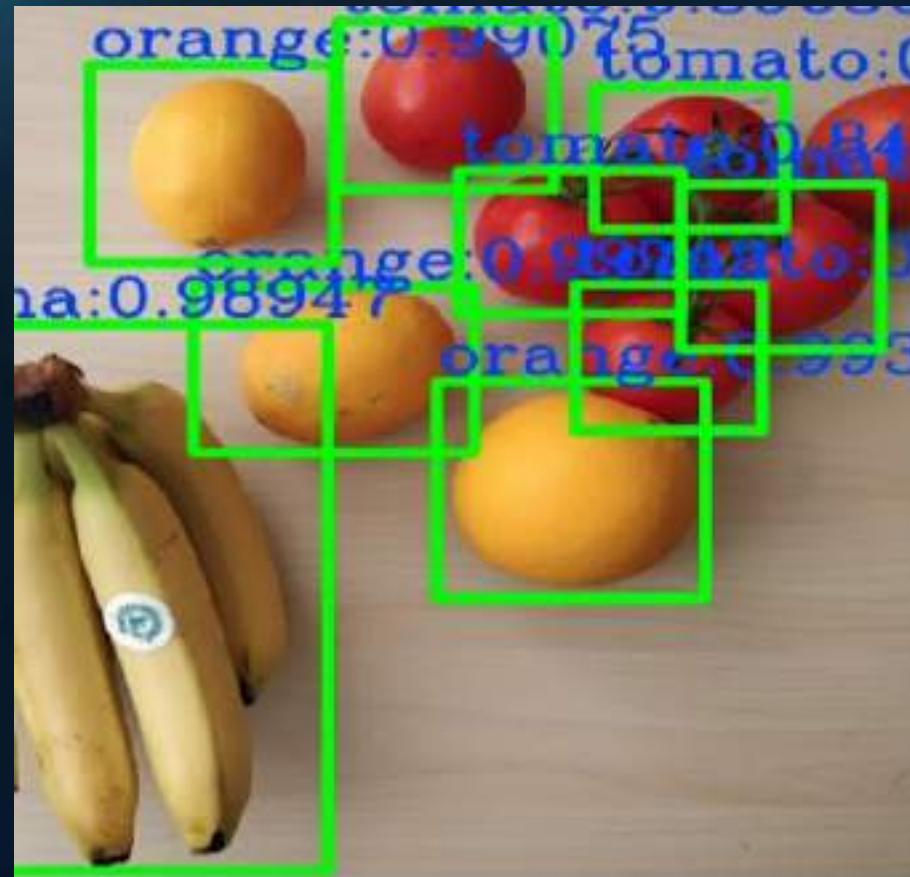


Crop and weed
Classification for
cultivating and Analysis

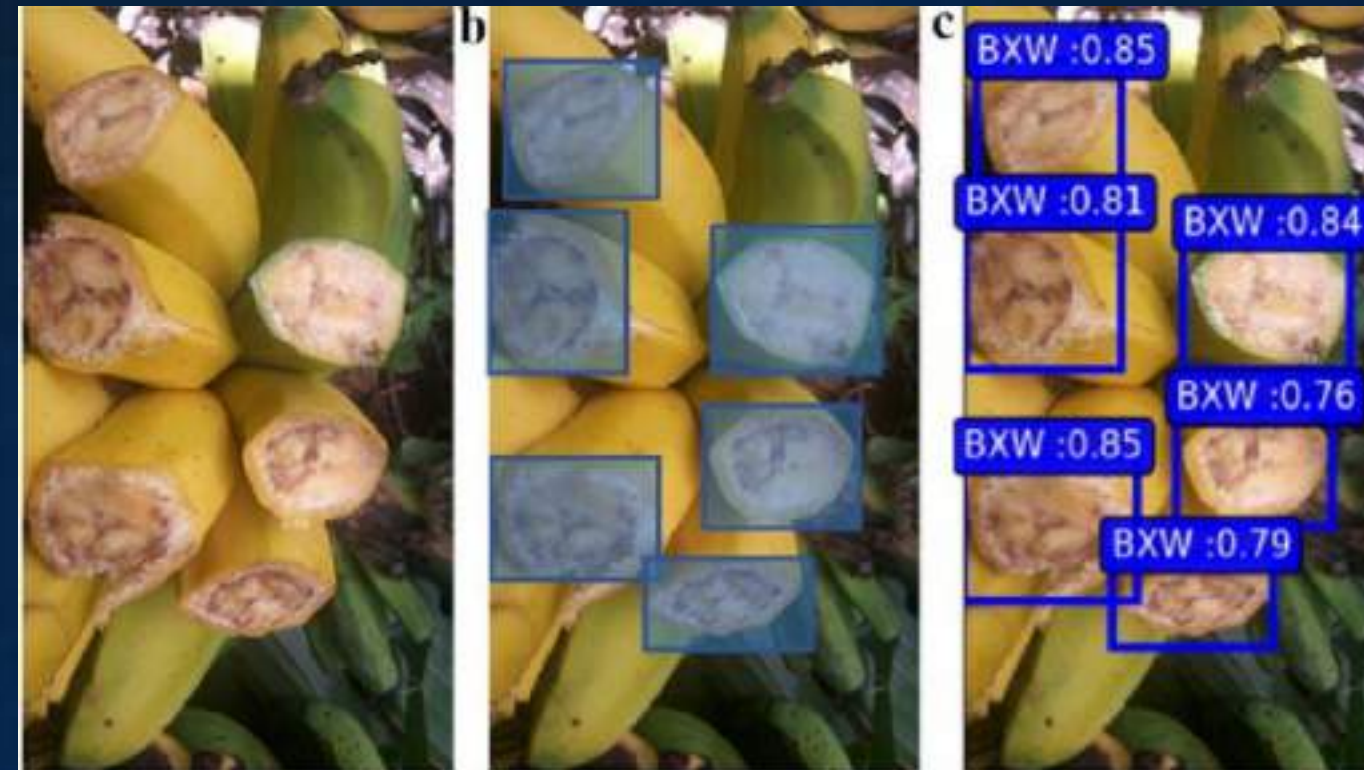
Helps us in stress and
strain analysis and also
among good and bad
crops

Helps us in analyzing the
crop disease and the
location where it is
occurring by dew
analysis and solution at
the fingertips

LOADING...



FRUIT AND CROP
ANALYSIS



FRUIT-CULTIVATING PERIOD
RECOGNITION; CONDITION OF
VEGETABLE ANALYSIS AND
SOLUTIONS SUGGESTION WITH
LOCATIONAL ACCESS



SUGGESTION OF SEASONAL
CROPS AND THE ANALYSIS OF
WHEN TO SUPPLY AND
MANIPULATE THE PRODUCTION
AT YOUR FINGERTIPS

BACK-BONE



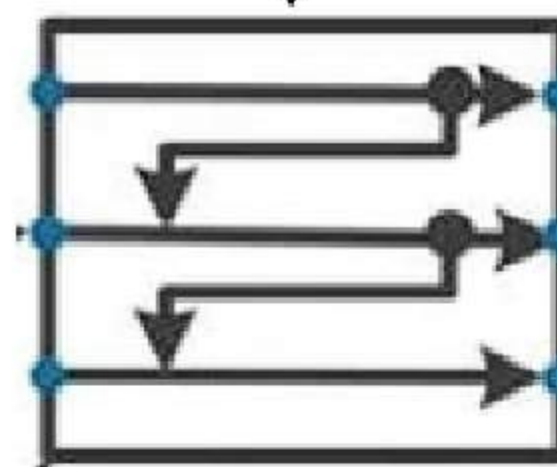
C-1

C-2

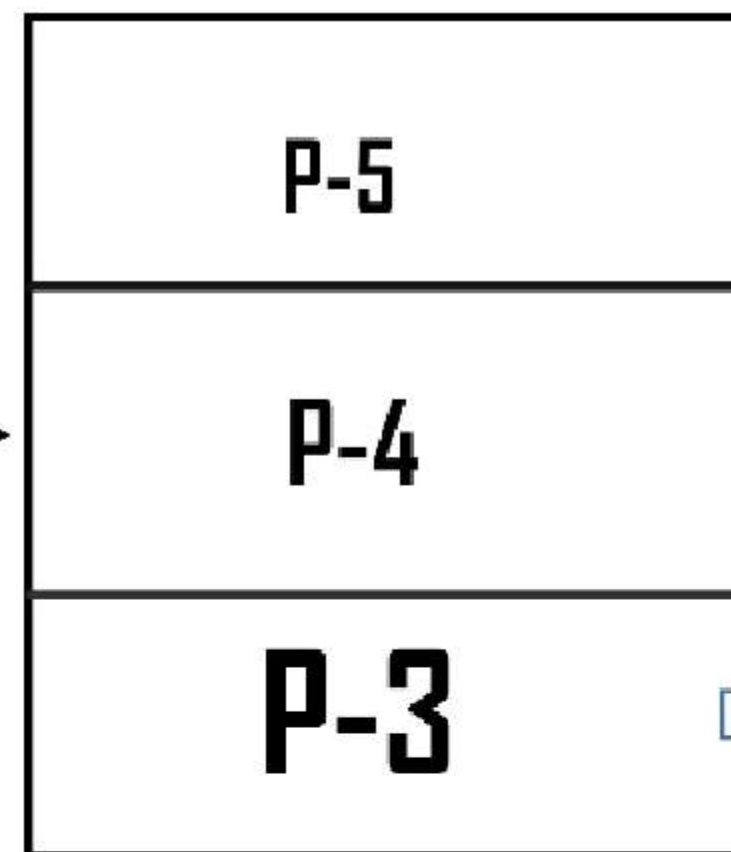
C-3

C-4

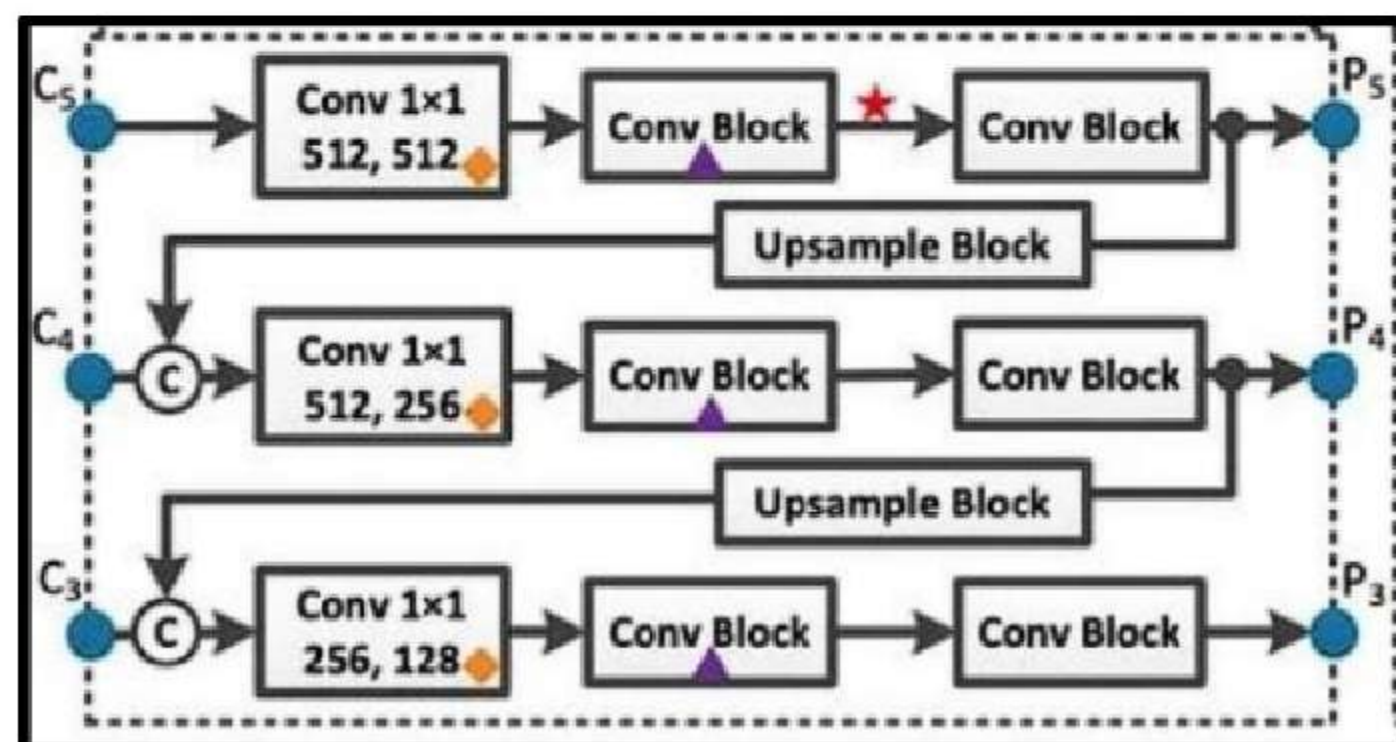
C-5



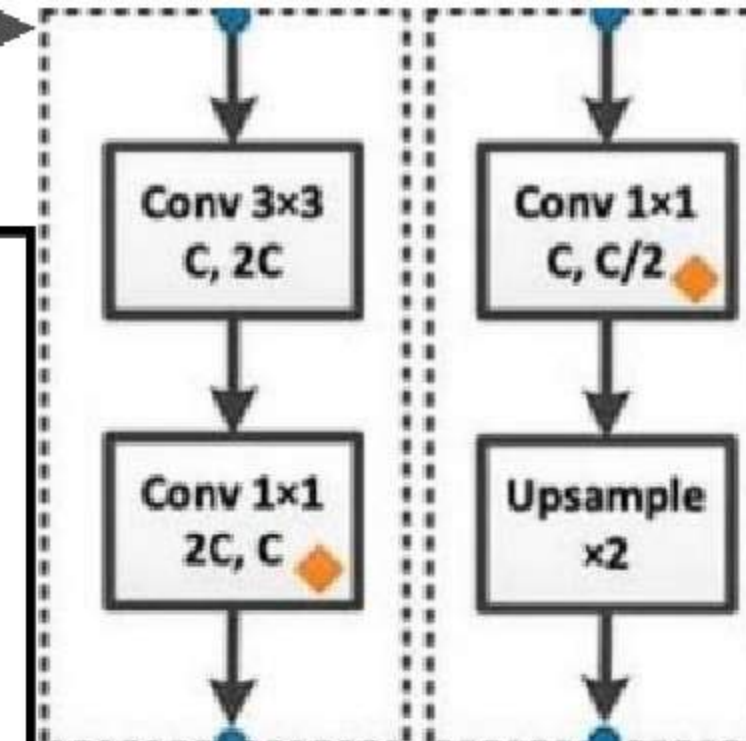
FPN



FEATURE PYRAMID

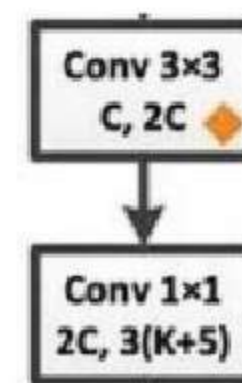
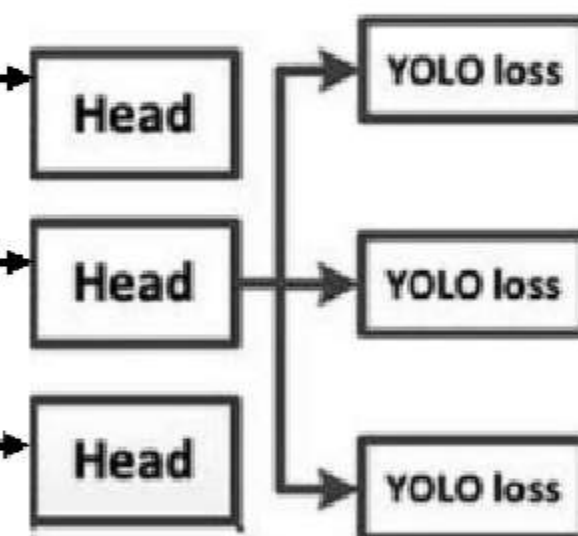


FPN

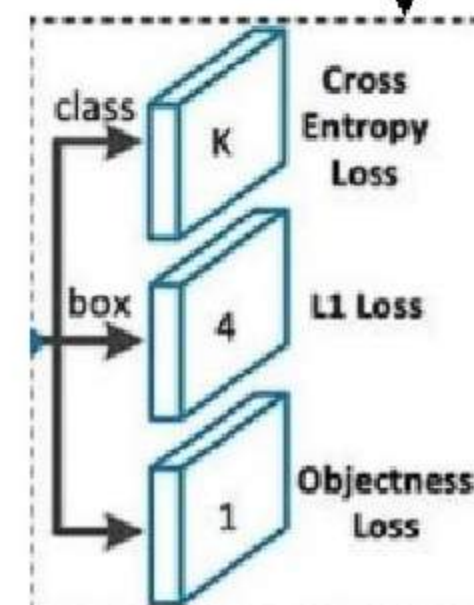


CONV BLOCK

UPSAMPLE BLOCK



HEAD





References:

1. Anurag, Dugar, Siuli Roy, and Somprakash Bandyopadhyay. "Agro- sense: Precision agriculture using sensor-based wireless mesh networks." In 2008 first itu-t kaleidoscope academic conference-innovations in ngn: Future network and services, pp. 383-388. IEEE, 2008.
2. Nandurkar, S. R., V. R. Thool, and R. C. Thool. "Design and develop- ment of precision agriculture system using wireless sensor network." In 2014 First international conference on automation, control, energy and systems (ACES), pp. 1-6. IEEE, 2014.
3. Sishodia, Rajendra P., Ram L. Ray, and Sudhir K. Singh. "Applications of remote sensing in precision agriculture: A review." Remote sensing 12, no. 19 (2020): 3136.
4. Kiani, Farzad, and Amir Seyyedabbasi. "Wireless sensor network and internet of things in precision agriculture." International Journal of Advanced Computer Science and Applications (2018).
5. Shafi, Uferah, Rafia Mumtaz, Jos´e Garc´ia-Nieto, Syed Ali Hassan, Syed Ali Raza Zaidi, and Naveed Iqbal. "Precision agriculture techniques and practices: From considerations to applications." Sensors 19, no. 17 (2019): 3796.
6. Mat, Ibrahim, Mohamed Rawidean Mohd Kassim, Ahmad Nizar Harun, and Ismail Mat Yusoff. "IoT in precision agriculture applications using wireless moisture sensor network." In 2016 IEEE Conference on Open Systems (ICOS), pp. 24-29. IEEE, 2016.
7. Delavarpour, N., Koparan, C., Nowatzki, J., Bajwa, S. and Sun, X., A technical study on UAV characteristics for precision agriculture applications and associated practical challenges. Remote Sensing, 13(6), p.1204, 2021.
8. Song, C., Zang, Y., Zhou, Z., Luo, X., Zhao, L., Ming, R., Zi, L. and Zang, Y., 2020. Test and comprehensive evaluation for the performance of UAV-based fertilizer spreaders. IEEE Access, 8, pp.202153-202163.
9. Rangappa, N., Prasad, Y.R.V. and Dubey, S.R., 2021. LEDNet: Deep learning-based ground sensor data monitoring system. IEEE Sensors Journal, 22(1), pp.842-850.
10. Sharma, R.P., Ramesh, D., Pal, P., Tripathi, S. and Kumar, C., 2021. IoT- enabled IEEE 802.15. 4 WSN monitoring infrastructure-driven fuzzy- logic-based crop pest prediction. IEEE Internet of Things Journal, 9(4), pp.3037-3045.
11. Ahmed, N., De, D. and Hussain, I., 2018. Internet of Things (IoT) for smart precision agriculture and farming in rural areas. IEEE Internet of Things Journal, 5(6), pp.4890-4899.
12. Tsouros, D.C., Bibi, S. and Sarigiannidis, P.G., 2019. A review on UAV- based applications for precision agriculture. Information, 10(11), p.349.
13. Ranjha, A., Kaddoum, G. and Dev, K., 2021. Facilitating URLLC in UAV-assisted relay systems with multiple-mobile robots for 6G net- works: A prospective of agriculture 4.0. IEEE Transactions on Industrial Informatics, 18(7), pp.4954-4965.



Thank You